PS-LPS Progress Report

Source path: `~/current_projects/geosmooth/split_handoffs/ps_lps_progress_2026-06-05/ps_lps_progress_report.tex`

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Purpose

This report records the current state of prediction-synchronized local polynomial smoothing, abbreviated *PS-LPS*. It is intended as a living progress report rather than a final method paper. The design document answers the question "what is PS-LPS?" This report answers:

> What has been implemented, what has been observed, what remains uncertain, and what should be audited or tested next?

The report is deliberately conservative. The first-batch results are encouraging, but they are preliminary until the implementation and experiment have passed an independent audit and have been tested on a broader validation suite.

1 Motivation

Ordinary local polynomial smoothing (LPS) fits one local polynomial model around each anchor point. This makes LPS simple and often effective, but it also means that neighboring local charts are fitted independently even though their supports overlap heavily. Two charts can make different predictions at the same overlap point, and ordinary LPS has no explicit mechanism to reduce that incoherence.

Before PS-LPS, we tested whether the sum of ordinary independent-chart local GCV scores could select global LPS parameters. That first experiment did not support using

$$\sum_i \mathrm{GCV}_i$$

as a standalone global selector for independent LPS. The diagnostic often favored high local degrees-of-freedom ratios and did not reliably track Truth RMSE on the first-batch examples.

PS-LPS takes a different route. Instead of using local GCV alone to select parameters, it modifies the smoother itself by adding a prediction synchronization term. The central idea is that overlapping local charts should not merely fit their own local responses; they should also agree where their prediction domains overlap.

2 Model Definition

Let $x_1, \ldots, x_n \in \mathbb{R}^D$ be observed covariate points and let y_i be the observed response at point x_i. For each anchor i, define a local support

$$N_i \subset \{1, \ldots, n\}, \qquad |N_i| = k.$$

Let $w_{ij} \geq 0$ be the kernel weight assigned by chart i to point $j \in N_i$. Let

$$\phi_i(x_j) \in \mathbb{R}^q$$

be the local polynomial feature vector for point x_j, expressed in the local chart centered at anchor i. The local coefficient vector at anchor i is $\beta_i \in \mathbb{R}^q$.

The local data-fit loss is

$$\mathcal{L}_{\text{local}}(\beta) = \frac{1}{2} \sum_{i=1}^{n} \sum_{j \in N_i} w_{ij} \{y_j - \phi_i(x_j)^\top \beta_i\}^2.$$

For two charts i and ℓ, define the overlap

$$O_{i\ell} = N_i \cap N_\ell.$$

The PS-LPS synchronization penalty is

$$\mathcal{S}(\beta) = \frac{1}{2} \sum_{(i,\ell)} \sum_{r \in O_{i\ell}} \omega_{i\ell r} \left\{ \phi_i(x_r)^\top \beta_i - \phi_\ell(x_r)^\top \beta_\ell \right\}^2.$$

The PS-LPS estimate is

$$\widehat{\beta} = \arg \min_{\beta_1, \ldots, \beta_n} \left[\mathcal{L}_{\text{local}}(\beta) + \lambda_{\text{sync}} \mathcal{S}(\beta) \right].$$

When $\lambda_{\text{sync}} = 0$, the model should reduce to independent LPS local chart fits with the same support, kernel, degree, and chart dimensions. This is an important audit gate for the implementation.

3 Overlap Weight Decision

The current recommended overlap weight is the normalized product of the two ordinary chart weights:

$$\omega_{i\ell r} = |O_{i\ell}| \frac{w_{ir} w_{\ell r}}{\sum_{s \in O_{i\ell}} w_{is} w_{\ell s} + \epsilon}.$$

The product $w_{ir} w_{\ell r}$ emphasizes points that are central in both charts. This matches the intended geometry: charts should be synchronized most strongly in the middle of their overlap, and less strongly near either chart's boundary. The pair normalization makes the overall synchronization scale more stable across chart pairs, kernels, and overlap sizes.

4 Implementation Status

The first implementation is an R reference prototype in

`~/current_projects/geosmooth/R/ps_lps.R`.

The exported user-facing function is

`fit.ps.lps()`.

The prototype currently:

- uses ordinary local PCA charts;

- supports fixed scalar chart dimension or a local chart-dimension vector;
- fits a fixed support size, kernel, degree, and chart-dimension policy;
- tunes only λ_{sync} by CV in the first experiment;
- assembles a sparse global least-squares system;
- solves sparse normal equations with a small scale-relative numerical ridge.

The current numerical ridge is a prototype stabilization device. It should be audited and eventually reported as a solver diagnostic, especially when testing whether $\lambda_{\text{sync}} = 0$ reproduces ordinary LPS.

5 Diagnostics

The PS-LPS implementation and first experiment record the following quantities:

- CV RMSE, computed on held-out responses;
- Observed RMSE, computed against the noisy observed response;
- Truth RMSE, computed against synthetic truth when available;
- synchronization energy $\mathcal{S}(\widehat{\beta})$;
- mean synchronization disagreement;
- local degrees-of-freedom ratios;
- total synchronized local GCV.

The synchronized local GCV diagnostic is defined after solving the synchronized problem. For chart i,

$$\mathrm{RSS}_i^{\mathrm{PS}} = \sum_{j \in N_i} w_{ij} \{y_j - \phi_i(x_j)^\top \widehat{\beta}_i\}^2.$$

Let r_i be the numerical rank of the local weighted design matrix. Then

$$\mathrm{GCV}_i^{\mathrm{PS}} = \frac{\mathrm{RSS}_i^{\mathrm{PS}}/k}{(1 - r_i/k)^2}, \qquad \mathrm{GCV}_\Sigma^{\mathrm{PS}} = \sum_i \mathrm{GCV}_i^{\mathrm{PS}}.$$

This is not the same as total local GCV for independent LPS charts. The residuals are computed from the synchronized chart coefficients $\widehat{\beta}_i$. This distinction matters because the preliminary results suggest that $\mathrm{GCV}_\Sigma^{\mathrm{PS}}$ may be more aligned with Truth RMSE than the independent-LPS version.

6 First-Batch Experiment

The first PS-LPS experiment used the frozen first-batch non-manifold datasets:

`~/current_projects/geosmooth/split_handoffs/`

`lps_local_auto_nonmanifold_first_batch_2026-06-05/`.

The generated PS-LPS report is:

`~/current_projects/geosmooth/split_handoffs/`

`ps_lps_first_batch_experiment_2026-06-05/`

`ps_lps_first_batch_experiment_report.html`.

The comparison included four methods:

- ordinary LPS with `chart.dim = "auto"`;
- ordinary LPS with `chart.dim = "local.auto"`;
- PS-LPS using the ordinary `auto` source settings;
- PS-LPS using the ordinary `local.auto` source settings.

In the first experiment, PS-LPS reused the corresponding ordinary LPS selected support size, kernel, degree, and chart-dimension policy, and tuned only

$$\lambda_{\text{sync}} \in \{0, 0.1, 1\}.$$

7 Audit Result and Response

The first implementation audit was blocked for two substantive reasons. First, the implementation used an unconditional ridge in the normal-equation solve, so $\lambda_{\text{sync}} = 0$ did not reproduce ordinary LPS fixed-chart fits. Second, the synchronization diagnostics were incorrectly reported as zero when $\lambda_{\text{sync}} = 0$, even though the unmultiplied disagreement $\mathcal{S}(\widehat{\beta})$ is generally nonzero.

The response was to make the ridge explicit rather than hidden. The current implementation has a user-facing parameter

$$\lambda_{\text{ridge}} \geq 0.$$

When $\lambda_{\text{ridge}} = 0$ and $\lambda_{\text{sync}} = 0$, PS-LPS is tested to reproduce ordinary independent LPS full-data fixed-chart fitted values. When $\lambda_{\text{ridge}} > 0$ and $\lambda_{\text{sync}} = 0$, the method is instead an independent ridge-stabilized LPS fit. Therefore any ridge-stabilized PS-LPS comparison must include the matched ridge-LPS baseline.

The diagnostics now compute and report the unmultiplied synchronization energy

$$\mathcal{S}(\widehat{\beta})$$

for every fitted coefficient vector whenever overlap rows exist, including the $\lambda_{\text{sync}} = 0$ case. The ambiguous diagnostic name `mean.sync.disagreement` has been retained only as a compatibility alias; the preferred diagnostic is

`mean.sync.squared.disagreement`.

8 Focused Tests

The focused PS-LPS test file is

`~/current_projects/geosmooth/tests/testthat/test-ps-lps.R`.

It currently checks:

- normalized product overlap weights have the intended per-pair mass;
- $\lambda_{\text{ridge}} = 0$ and $\lambda_{\text{sync}} = 0$ reproduce ordinary LPS fixed-chart fitted values;
- positive ridge with $\lambda_{\text{sync}} = 0$ reproduces the independent ridge-LPS solver path;
- synchronization energy and mean squared disagreement are nonzero diagnostics at $\lambda_{\text{sync}} = 0$ when charts disagree.

The targeted test run passed with 9 expectations and no failures.

9 Refined First-Batch Experiment

The refined report is:

`~/current_projects/geosmooth/split_handoffs/`

`ps_lps_first_batch_refined_experiment_2026-06-05/`

`ps_lps_first_batch_refined_experiment_report.html`.

The refined comparison separates six methods:

- ordinary LPS `auto`;
- ridge-LPS `auto`;
- PS-LPS `auto`;
- ordinary LPS `local.auto`;
- ridge-LPS `local.auto`;
- PS-LPS `local.auto`.

All ridge-LPS and PS-LPS rows use

$$\lambda_{\text{ridge}} = 10^{-8}.$$

The PS-LPS grid remains

$$\lambda_{\text{sync}} \in \{0, 0.1, 1\},$$

and ridge-LPS is the matched $\lambda_{\text{sync}} = 0$ baseline.

All 84 method–dataset rows completed successfully. The best Truth-RMSE method counts over the 14 first-batch datasets were:

Method	Number of datasets where best
PS-LPS local.auto	8
PS-LPS auto	5
ridge-LPS local.auto	1
ordinary LPS variants	0
ridge-LPS auto	0

Median Truth RMSE by method was:

Method	Median Truth RMSE
PS-LPS local.auto	0.04048
PS-LPS auto	0.04517
ridge-LPS local.auto	0.04868
LPS local.auto	0.05755
ridge-LPS auto	0.05752
LPS auto	0.06978

The refined delta decomposition is the main interpretive result. The median Truth-RMSE changes were:

Method	Reference	Median Truth-RMSE delta
ridge-LPS auto	ordinary LPS auto	−0.00651
ridge-LPS local.auto	ordinary LPS local.auto	−0.00638
PS-LPS auto	ordinary LPS auto	−0.02437
PS-LPS local.auto	ordinary LPS local.auto	−0.02190
PS-LPS auto	ridge-LPS auto	−0.01048
PS-LPS local.auto	ridge-LPS local.auto	−0.00865

Thus the initial favorable result was partly a ridge-stabilization effect, but not only a ridge effect. In this refined first-batch experiment, positive prediction synchronization still improved over the matched ridge-LPS baseline in median Truth RMSE. The dataset-wise caveat is that each PS-LPS variant improved over its matched ridge-LPS baseline on 13 of 14 datasets, not on every dataset. Therefore the appropriate short summary is "median improvement over matched ridge-LPS, with 13/14 dataset-wise wins for each PS variant."

10 Open Questions

The following questions remain open and should guide the next phase:

- Is the improvement over ridge-LPS stable under a wider λ_{sync} grid?
- How sensitive are the conclusions to the ridge scale λ_{ridge}?
- Does PS-LPS still improve on prospective datasets that were not used to motivate the method?
- Does total synchronized local GCV remain aligned with CV RMSE and Truth RMSE on larger suites?
- Should PS-LPS select support size, kernel, degree, and chart dimension directly, or remain a synchronization refinement around an ordinary LPS-selected candidate?

- Is the R prototype sufficient for near-term experiments, or should the sparse system assembly and solve be moved to C++?

11 PS-LPS-S1 Sensitivity Plan

The re-audit accepted the refined ridge-stabilized contract with minor follow-ups. The next validation phase is therefore not yet a prospective experiment. The immediate phase is:

PS-LPS-S1: first-batch lambda/ridge sensitivity.

Its purpose is to test whether the refined first-batch PS-LPS signal is stable to the two prototype tuning axes:

$$\lambda_{\text{sync}} \quad \text{and} \quad \lambda_{\text{ridge}}.$$

The planned synchronization grid is

$$\lambda_{\text{sync}} \in \{0,\ 0.01,\ 0.03,\ 0.1,\ 0.3,\ 1,\ 3,\ 10\},$$

and the planned ridge grid is

$$\lambda_{\text{ridge}} \in \{0,\ 10^{-10},\ 10^{-8},\ 10^{-6}\}.$$

For every ridge scale, the matched baseline is the candidate with $\lambda_{\text{sync}} = 0$ and the same support size, kernel, degree, chart rule, and ridge scale. When $\lambda_{\text{ridge}} = 0$, this is ordinary unregularized LPS. When $\lambda_{\text{ridge}} > 0$, this is ridge-LPS.

The S1 report should include:

- selected λ_{sync} by dataset, chart rule, and ridge scale;
- candidate-level Truth RMSE over the $(\lambda_{\text{ridge}}, \lambda_{\text{sync}})$ grid;
- selected Truth-RMSE deltas against the matched $\lambda_{\text{sync}} = 0$ baseline;
- median and dataset-wise win summaries;
- total local GCV, synchronization energy, and mean squared overlap disagreement diagnostics.

The decision criterion is conservative: proceed to prospective PS-LPS experiments only if the improvement over matched ridge-LPS is not confined to a single ridge scale or to a boundary value of λ_{sync}.

12 PS-LPS-S1 Results

The S1 experiment has now been run on the frozen first-batch non-manifold suite. The report is:

`/current_projects/geosmooth/split_handoffs/`
`ps_lps_s1_lambda_ridge_sensitivity_2026-06-05/`
`ps_lps_s1_lambda_ridge_sensitivity_report.html`.

The run evaluated 896 candidate rows:

$$14 \text{ datasets} \times 2 \text{ chart rules} \times 4\ \lambda_{\text{ridge}} \text{ values} \times 8\ \lambda_{\text{sync}} \text{ values}.$$

All candidate rows completed with status `ok`. The requested $\lambda_{\text{ridge}} = 0$ rows are included, but the numerical solver did use its internal fallback ridge on a small number of zero-ridge candidate solves. The S1 tables record the realized λ_{ridge} diagnostics as `ridge_median` and `ridge_max`. Thus the zero-ridge rows are valuable as a stress test of the no-ridge path, but they should not be described as uniformly exact unregularized normal-equation solves.

The selected-by-CV median Truth-RMSE deltas against the matched $\lambda_{\text{sync}} = 0$ baseline were:

Chart rule	λ_{ridge}	Median delta	Wins
`auto`	0	−0.02224	12/14
`local.auto`	0	−0.01523	13/14
`auto`	10^{-10}	−0.01319	12/14
`local.auto`	10^{-10}	−0.01104	12/14
`auto`	10^{-8}	−0.01523	12/14
`local.auto`	10^{-8}	−0.01095	12/14
`auto`	10^{-6}	−0.01425	12/14
`local.auto`	10^{-6}	−0.01057	12/14

Negative deltas mean that the selected synchronized fit has lower Truth RMSE than the matched independent baseline at the same ridge scale. The signal is therefore not confined to one ridge scale: every ridge scale and both chart rules show a negative median delta.

The caution is that the selected synchronization strength is usually at the upper boundary of the S1 grid. Across all ridge scales, the median selected λ_{sync} was 10 for both chart rules. This does not invalidate the positive S1 result, but it does mean that the next sensitivity run should extend the synchronization grid above 10 and should examine whether the Truth-RMSE profile plateaus, improves further, or eventually deteriorates. In short, S1 supports prediction synchronization as a useful refinement, but it does not yet freeze the synchronization search range.

Summary

PS-LPS has moved from a concept note to a tested ridge-stabilized prototype. The blocked audit identified a real interpretability problem, and the refined experiment now separates ordinary LPS, ridge-LPS, and synchronized PS-LPS. The first-batch evidence remains encouraging because PS-LPS improves over the matched independent baseline in median Truth RMSE across the tested ridge scales. The main limitation is now different: S1 frequently selects the largest tested synchronization weight, so the next experiment should extend λ_{sync} before prospective validation.